

Predicting TCR Cross-Reactivity Before It Reaches a Patient

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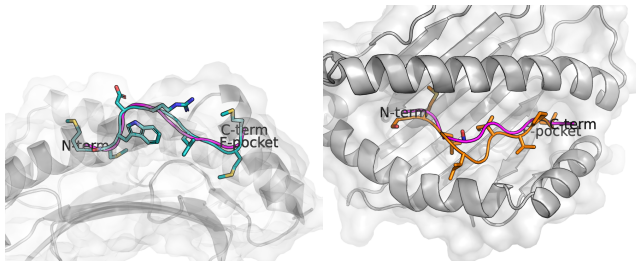


Figure 1: Proposed framework. A two-state discrimination test in a cross-reactive pMHC–TCR system. Diffusion models sample from the cognate conformation (teal, left) toward a known off-target conformation (orange, right), two distinct peptides the same receptor engages. Each generated backbone (magenta) is scored against both, and credited only if substantially closer to the one it is claimed to match.

The problem. Engineered T-cell receptor therapies promise a transformative future for cancer treatment, and with the 2024 approval of afamitresgene autoleucel that future has begun to arrive¹. Cross-reactivity stands between that beginning and routine use. An affinity-enhanced MAGE-A3 receptor that had cleared extensive preclinical testing killed the first two patients who received it, by cross-recognizing a self-peptide from the muscle protein titin sharing five of nine residues with the intended target². That peptide was identified only after both deaths³.

Why the field is stuck. The bottleneck has never been imagination, it has been throughput. The space of peptides a human cell can present is on the order of 10^{11} , so exhaustive experimental testing of one receptor against it is not merely expensive but arithmetically out of reach, while the aggressive tumors these therapies target leave a clinical window measured in weeks. HLA polymorphism compounds this. Measured binding data are sparse for most alleles and sparsely for underrepresented populations, precisely the patients personalized therapy is meant to reach, so the predictors that might substitute for experiment are weakest where they are needed most. Positional scanning mutagenesis, the standard alternative, resolves one receptor against one substitution at a time and does not scale to a space combinatorial in both peptide sequence and receptor identity.

The approach. The tools to test this at scale have only now come within reach. I would bring two of my own design. ESMCBA is a binding-affinity predictor built on a protein language model with domain adaptation, transferring signal from data-rich to data-scarce HLA alleles and extending coverage into the allelic space conventional predictors abandon⁴. The second is a structure-guided generative frame-

work in which diffusion models, conditioned on the geometry of the MHC groove, propose peptides compatible with a given presentation rather than enumerating them blindly⁵. Inverse-folding models supply the third component, reading a fixed backbone and decoding which residues that geometry admits⁶, which localizes the positions a receptor reads. Together they generate a candidate off-target set, rank it, and localize it (Fig. 1). The instrument is what converts my two tools from proofs of concept into a working safeguard.

How it is proven. Each prediction is falsifiable at the bench. The receptor either engages the flagged peptide or it does not. Validation proceeds in two stages. Retrospectively, the screen runs blind against receptors whose off-targets are already documented and must recover them. The MAGE-A3 case is the primary test, since the titin epitope is known and was missed by the methods in use at the time. A screen that fails to flag it fails outright. Prospectively, top-ranked candidates for a panel of characterized receptors go to binding measurement and, for those that bind, to functional assay. This yields a quantity the field currently lacks, a false-negative rate for cross-reactivity prediction on peptides never seen in training. Both stages apply the same two-state discrimination criterion, so a candidate counts only when it sits closer to the off-target conformation than to the cognate one, never on generic proximity to the groove. The validation loop is short enough to close within the appointment, not left to a successor.

What becomes possible. A cross-reactivity screen that runs *in silico* before infusion turns a lethal surprise into a routine preclinical filter, applied to every candidate receptor rather than the few that fit an experimental budget, and extended to the allele diversity of real patient populations rather than the handful with deep binding data. The same framework generalizes past T-cell receptors to any therapy built on engineered molecular recognition, where the failure mode is likewise specificity rather than potency. The societal stake is measured in patients, not benchmarks.

Trajectory. The first problem is chosen because it is winnable inside a year, the second because it is worth a career. Near term the deliverable is the validated screen and its false-negative rate. Beyond it lies the harder question the same instrument opens, whether specificity can be designed at the outset rather than audited afterward, so that receptors are built against the off-target landscape instead of tested against it.

Given an instrument of this reach, this is the first thing I would point it at.

¹N. Singh. Approval of the first TCR-based cell therapy. *Mol. Ther.* 32:3195, 2024.

²G. P. Linette et al. Cardiovascular toxicity and titin cross-reactivity of affinity-enhanced T cells in myeloma and melanoma. *Blood* 122:863–871, 2013.

³B. J. Cameron et al. Identification of a titin-derived HLA-A1-presented peptide as a cross-reactive target for engineered MAGE-A3-directed T cells. *Sci. Transl. Med.* 5:197ra103, 2013.

⁴S. Mares, A. Espinoza, N. M. Ioannidis. Continued domain-specific pre-training of protein language models for pMHC-I binding prediction. *PMLR* 311:304–325, 2025.

⁵S. Mares, A. Espinoza, N. M. Ioannidis. Generation of structure-guided pMHC-I libraries using diffusion models. *ICML GenAI and Biology Workshop*, 2025.

⁶H. V. Ribeiro-Filho et al. Exploring the potential of structure-based deep learning approaches for T cell receptor design. *PLOS Comput. Biol.*, 2024.